

PROJECT EXPO 2026

Smart IV Drip Monitoring Using IOT

TEAM MEMBERS:

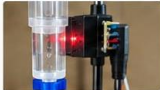
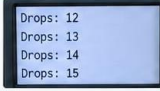
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TESTING & VALIDATION REVIEW(TVR):

FUNCTIONAL TESTING REPORT:

The developed Smart IV Drip Monitoring System was tested to verify the operation of all hardware and software modules. The TCRT5000 IR sensor successfully detected the IV drops passing through the drip chamber. The ESP32 processed the sensor data, calculated the flow rate, and transmitted the information through Wi-Fi. The SG90 servo motor responded to control commands and regulated the IV tube flow. The complete system operated successfully during testing.

Module / Function	Test Setup / Image	Expected Result	Actual Result	Status
 Drop Detection	 IV chamber with TCRT5000 IR sensor	Detect every IV drop passing through the chamber.	Drops were detected accurately for each drop.	✓ Pass
 Drop Counting	 Drops: 12 Drops: 13 Drops: 14 Drops: 15	Count drops accurately.	Drop count matched with actual drops.	✓ Pass
 Flow Rate Calculation	 Flow Rate: 8.0 ml/min Drops: 12	Calculate flow rate (ml/min).	Flow rate displayed correctly (e.g., 8.0 ml/min).	✓ Pass
 Servo Control	 SG90 servo motor compressing the tube	Regulate / stop IV flow when commanded.	Servo moved to required angle and stopped/regulated the flow successfully.	✓ Pass
 Wi-Fi Communication	 Blynk IV Monitoring Drop Count: 58 Flow Rate: 9.2 ml/min Connected	Send data to dashboard in real-time.	Data updated on Blynk in real-time (e.g., every 2 seconds).	✓ Pass
 Remote Monitoring	 Blynk IV Monitoring Drop Count: 58 Flow Rate: 9.2 ml/min Connected	View data remotely on mobile/web.	Dashboard accessible from mobile device and laptop.	✓ Pass

PERFORMANCE EVALUATION:



CONCLUSION:

The Smart IV Drip Monitoring System demonstrated reliable drop detection, accurate flow-rate estimation, stable Wi-Fi communication, and effective servo-based flow regulation. The system maintained continuous operation for one hour without significant performance degradation.

TEST CASES:

Test Case 1:

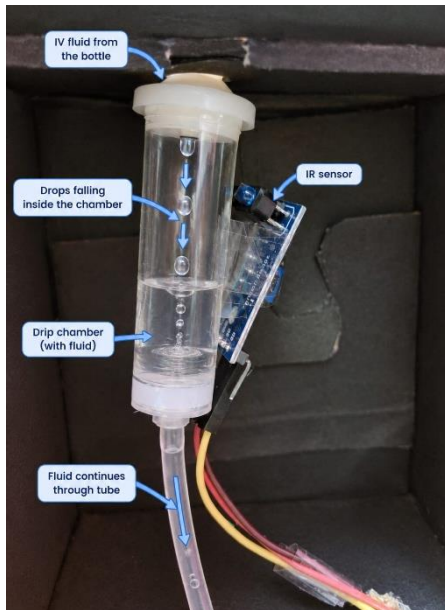
Objective: Detect IV drops

Input: Water flowing through chamber

Expected: Drop count increases

Result: Pass

Fig: Drop Detection



Test Case 2:

Objective: Stop flow using servo

Input: Servo command

Expected: Tube compressed and flow stops

Result: Pass

Fig: Stop flow using servo



Test Case 3:

Objective: Wi-Fi monitoring

Input: Connect ESP32 to Blynk

Expected: Data visible remotely

Result: Pass

Fig: Wi-Fi monitoring



ERROR ANALYSIS:

S.No	Error Observed	Cause Identified	Corrective Action	Result
1	 IR sensor not detecting drops	Sensor and drip chamber misalignment	Repositioned sensor and adjusted sensitivity	Detection improved
2	 False drop counting	Ambient light interference	Covered sensor sides and reduced external light	Accurate counting achieved
3	 Missed drop detection	Drops not passing through sensing area	Adjusted sensor placement and alignment	Consistent detection obtained
4	 Servo not stopping flow completely	Insufficient tube compression	Added rubber support and optimized servo angle	Better flow control achieved
5	 Wi-Fi disconnection	Weak network signal	Implemented automatic reconnection	Stable communication established
6	 Delay in dashboard updates	Network latency	Optimized data transmission interval	Faster updates observed
7	 Unstable flow-rate readings	Variation in drop size and speed	Applied averaging/filtering algorithm	Stable readings achieved
8	 ESP32 restarting unexpectedly	Power supply fluctuation	Used regulated 5V power source	Stable operation achieved

CONCLUSION:

The identified errors were successfully analysed and resolved through proper debugging and system optimization. The final prototype achieved improved accuracy, stable communication, reliable servo operation and consistent performance during testing.

DEBUGGING REPORT:

1. Issue: IR sensor not detecting drops

Action: Checked wiring, repositioned TCRT5000 sensor, adjusted sensitivity

Outcome: Drop detection improved

2. Issue: False drop counting

Action: Reduced ambient light interference and recalibrated sensor

Outcome: Accurate counting achieved

3. Issue: Missed drop detection

Action: Realigned sensor with drip path

Outcome: Consistent detection obtained

4. Issue: Servo not stopping flow completely

Action: Modified servo mounting and added rubber support

Outcome: Better tube compression achieved

5. Issue: ESP32 not connecting to Wi-Fi

Action: Verified SSID/password and restarted connection

Outcome: Successful Wi-Fi connection established

6. Issue: Dashboard not updating data

Action: Checked Blynk configuration and communication code

Outcome: Real-time monitoring restored

7. Issue: Unstable flow-rate values

Action: Applied averaging/filtering algorithm

Outcome: Stable readings obtained

8. Issue: ESP32 restarting unexpectedly
Action: Used regulated 5V power supply
Outcome: Stable operation achieved

CONCLUSION:

After debugging, the prototype achieved reliable drop detection, stable IoT communication, accurate flow-rate monitoring, and effective servo control.

POWER CONSUMPTION ANALYSIS:

1. Objective:

To estimate the total power consumed by the Smart IV Drip Monitoring System during standard operation and evaluate its suitability for continuous, long-term medical monitoring applications.

2. Power Consumption of Components:

The hardware architecture integrates a microcontroller unit for Wi-Fi telemetry, an optical sensor for drop detection, and a precision actuator for flow control. Detailed specifications are outlined below:

Component	Operating Voltage	Current Consumption
ESP32 Microcontroller	5V	80 mA
TCRT5000 IR Sensor	3.3V	15 mA
SG90 Servo Motor	5V	250 mA (maximum)
Total System Load	5V	345 mA

3. Power Calculation & Results:

Total power consumption is determined using Ohm's Law and standard electrical power equations:

$$\text{Formula: Power (P) = Voltage (V) } \times \text{ Current (I)}$$

$$\text{Calculation: } P = 5V \times 0.345A = 1.725 \text{ W}$$

Parameter	Calculated Value
Supply Voltage	5V
Total Current Consumption	345 mA (0.345 A)
Total Power Consumption	1.725 W
System Category	Low-Power IoT Device

4. Conclusion:

The Smart IV Drip Monitoring System consumes approximately 1.725 W under peak operational load. Because of this low power requirement, the system can be reliably powered using standard 5V USB wall adapters or portable USB power banks. This portability and energy efficiency make it exceptionally well-suited for continuous inpatient monitoring and remote healthcare settings.

RELIABILITY TESTING:

1.Test Procedure:

1. The IV drip monitoring hardware setup was powered and operated continuously under standard testing conditions.
2. The TCRT5000 IR optical sensor continuously monitored and registered fluid drop counts.

- 3. The ESP32 microcontroller dynamically calculated the flow rate and transmitted telemetric data over Wi-Fi to the central dashboard.
- 4. The SG90 servo motor actuator was triggered through multiple open/close cycles to regulate drip flow rate.
- 5. System stability, communication connectivity, and hardware responsiveness were recorded over an extended 60-minute evaluation period.

2. Reliability Testing Results:

Test Parameter	Test Condition	Observation	Result
Continuous Operation	System operated for 30 minutes	No interruption in operation	Pass
Long Duration Test	System operated for 60 minutes	Stable performance maintained	Pass
Drop Detection Reliability	Continuous drip flow	Drops detected consistently	Pass
Servo Reliability	Multiple open/close operations	Servo responded correctly	Pass
Wi-Fi Communication Reliability	Continuous data transmission	Data updated successfully on dashboard	Pass
ESP32 Stability	Extended operation	No unexpected restart observed	Pass

3. Reliability Testing Metrics Summary:

Reliability Metric	Observed Result
Operating Time	60 Minutes
System Failures	0
ESP32 Restarts	0

Wi-Fi Failures	0
Overall Reliability	Good / Excellent

4. Conclusion:

The Smart IV Drip Monitoring System demonstrated exceptionally reliable performance throughout all test scenarios. The system successfully maintained continuous drop detection, accurate flow-rate calculations, dependable IoT telemetry transmission, and responsive servo control over the 60-minute trial without a single failure or restart. These results confirm the system's robustness and suitability for continuous long-duration clinical and remote healthcare monitoring applications.

FUTURE IMPROVEMENT:

1. Future Enhancements:

S.No	Improvement	Benefit
1	Use medical-grade flow sensors	Higher accuracy in flow-rate measurement
2	Mobile app notifications	Instant alerts to nurses and caregivers
3	Cloud data storage	Remote access and patient record maintenance
4	Battery backup system	Continuous operation during power failure
5	Multi-patient monitoring	Monitor several IV setups from a single dashboard

S.No	Improvement	Benefit
6	AI-based anomaly detection	Early identification of abnormal flow conditions
7	Automatic IV bottle level monitoring	Timely replacement of IV fluids
8	Hospital database integration	Centralized patient monitoring and management

2. Future Scope:

- Improve flow-rate accuracy using advanced medical sensors.
- Develop a dedicated mobile application for healthcare staff.
- Enable cloud-based monitoring from anywhere.
- Add battery backup for uninterrupted operation.
- Expand the system to monitor multiple patients simultaneously.
- Incorporate intelligent analytics for predictive healthcare monitoring.